

# SCI-MAP

## Ordinary Mapmaking and Observation Coaddition

### Formal Scientific/Engineering Contract

Scientific contract version: **v0.1**  
 Document revision: **r0.4**  
 Status: **Author draft for scientific-owner review**  
 Scientific owner: Grant Wilson  
 Author-packet approval: 2026-08-16

**Authority boundary.** This is the normative formal view of the implementation-independent scientific contract. It preserves the canonical equations, 52 requirements, 25 predictions, exact state and support semantics, provenance, and conformance routing. The separate science-team rationale r0.4 explains the same science without replacing this authority. This document does not claim implementation conformity, representation fidelity, scientific validation, observational performance, or production readiness.

## Physical model and contract at a glance

MAP consumes the realized MAP-facing output of PTC; it never bypasses PTC to consume CAL output directly. ALIGN/AST supplies the coordinate facts and MAP supplies the target grid and declared projection. SCI-MAP applies one fixed, positive-coefficient linear averaging operator to the signal and to every declared linear companion. A coadd then applies a second positive-coefficient operator to atomically admitted, compatible observation bundles on the same grid. All scientific statements are conditional on the complete admitted calibration, alignment, eligibility, coefficient, response, geometry, and covariance state.

| Facet    | Compact authority                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input    | Realized SCI-PTC MAP-facing Stokes-I product in <b>mJy/beam</b> , with PTC-owned identity, availability, producer-local validity, signal, and coefficient facts; ALIGN/AST-owned coordinates and coordinate validity; and MAP-owned target-grid, projection, boundary, support, and use policy.                                                         |
| Output   | A complete immutable observation-map bundle and, when requested and admissible, a complete centered-integer common-grid observation coadd bundle.                                                                                                                                                                                                       |
| Equation | On admitted membership, $N_p = \sum_i a_{pi} x_i$ and $Q_p = \sum_i a_{pi}$ . The normalized scientific vector contains exactly the rows authorized by the effective support policy, with $\hat{x}_p = N_p/Q_p$ there; response and covariance use the same explicit row selection. Coadd repeats this restricted structure over complete observations. |
| Source   | Owner-approved Scope Brief and prior frozen core, amended by the owner-authorized PTC-to-MAP handoff correction recorded in <b>CHANGE_LOG_R0.4.md</b> .                                                                                                                                                                                                 |
| Status   | Contract author draft. Algebraic predictions are preregistered below; none has been executed here. All stronger claim layers remain unassessed.                                                                                                                                                                                                         |

## Contents

|                                                          |          |
|----------------------------------------------------------|----------|
| <b>Physical model and contract at a glance</b>           | <b>1</b> |
| <b>1 Scientific boundary and estimand</b>                | <b>3</b> |
| <b>2 Linear operator, response, and absolute mode</b>    | <b>4</b> |
| <b>3 Input classes and falsifiable response behavior</b> | <b>5</b> |
| <b>4 Conditional uncertainty and statistical labels</b>  | <b>6</b> |

|    |                                                       |    |
|----|-------------------------------------------------------|----|
| 5  | Eight facts, support, and raw validity                | 7  |
| 6  | Observation coaddition on one common grid             | 8  |
| 7  | Identity, WCS, units, and lifecycle                   | 9  |
| 8  | Failure, downstream boundary, and immutable parentage | 10 |
| 9  | Uncertainty, validity, validation, and claim layers   | 11 |
| 10 | Open owner decisions and scientific disposition       | 12 |
| A  | Owner-decision register                               | 13 |
| B  | Requirement and crosswalk summary                     | 15 |
| C  | Canonical requirements                                | 16 |
| D  | Falsifiable predictions                               | 21 |
| E  | Product vocabulary                                    | 23 |
| F  | Admitted source register and independence statement   | 24 |

# 1 Scientific boundary and estimand

The ordinary observation map estimates the coefficient-weighted average of the admitted realized PTC sample quantity in each declared map pixel. Its meaning is inherited through the required CAL-to-PTC route, from ALIGN/AST's coordinate relation, and from PTC's transformed signal, product identity, availability, producer-local validity, and MAP-facing coefficient. SCI-MAP does not repair or reinterpret those facts. VAL may register and evaluate a named rule, but it is not the author of producer or MAP policy.

Let  $i = (t, d)$  identify a candidate sample occurrence and  $p$  a pixel. Here  $x_i$  is the realized PTC transformed signal and  $\omega_i$  is its PTC-supplied MAP-facing analysis coefficient. Neither is the paired upstream  $x/r$  detector-component coordinate, and MAP does not reinterpret  $r$  as a response. The effective positive coefficient  $a_{pi} = G_{pi}\omega_i$  combines MAP's declared nonnegative projection coefficient with that admitted analysis coefficient. The fixed contribution set  $\mathcal{C}_p$  requires the source-declared PTC output to be available at its own declared scope and every MAP-owned admission condition to pass before arithmetic. No universal occurrence-to-product or product-to-occurrence failure hierarchy is inferred. Let  $\Pi_{\text{eff}}$  be the recorded effective product support policy and  $\mathcal{S}_{\text{out}}(\Pi_{\text{eff}})$  the exact set of map rows for which every support predicate that policy designates applicable passes. This is a row domain, not a numerical mask. The frozen ordinary result is then reused directly:

$$a_{pi} = G_{pi} \omega_i, \tag{1}$$

$$\mathcal{C}_p = \{i : \text{available}_{\text{PTC}}(i) = 1, \text{ admit}_{\Pi_{\text{MAP}}}(i, p) = 1, a_{pi} > 0\}, \tag{2}$$

$$\mathcal{S}_{\text{out}}(\Pi_{\text{eff}}) = \{p : \text{authorize}_{\Pi_{\text{eff}}}(p) = 1\}, \tag{3}$$

$$N_p = \sum_{i \in \mathcal{C}_p} a_{pi} x_i, \quad Q_p = \sum_{i \in \mathcal{C}_p} a_{pi}, \tag{4}$$

$$K_p = \sum_{i \in \mathcal{C}_p} a_{pi} k_i, \tag{5}$$

$$\hat{x}_p = N_p / Q_p, \quad \hat{k}_p = K_p / Q_p, \quad p \in \mathcal{S}_{\text{out}}, \tag{6}$$

The supported division is not a recipe for filling unsupported pixels. Raw accumulators and representation-specific storage slots may exist outside  $\mathcal{S}_{\text{out}}$ , but the normalized scientific vector has no row there. A dense zero, sentinel, or coincidentally finite payload is not a semantic substitute and is not measured zero sky. Likewise,  $Q_p$  is a normalization fact. Its dimensions or historical name do not make it precision.

The active quantity boundary is **mJy/beam**. A beam-calibrated sample and map require the admitted upstream beam and calibration identities; SCI-MAP does not supply those facts. Accepted tokens for other units do not authorize conversion. V0.1 is Stokes I only and does not authorize measured-R, Q/U, JINC, maximum-likelihood mapmaking, mosaicking, or reprojection.

## 2 Linear operator, response, and absolute mode

For selected samples  $\mathbf{y}$ , projection matrix  $G$ , diagonal coefficient matrix  $\Omega$ , forward sky-to-sample operator  $H$ , and exact effective output-row selector  $J_{\text{out}}$ , the map operator and its conditional response and covariance are:

$$D_Q = \text{diag}(G\Omega\mathbf{1}), \quad J_{\text{out}}\mathbf{z} = (z_p)_{p \in \mathcal{S}_{\text{out}}}, \quad D_{Q,\text{out}} = J_{\text{out}}D_QJ_{\text{out}}^\top, \quad (7)$$

$$A_{\text{out}} = D_{Q,\text{out}}^{-1}J_{\text{out}}G\Omega, \quad \hat{\mathbf{x}}_{\text{out}} = A_{\text{out}}\mathbf{y}, \quad (8)$$

$$R_{\text{out}} = A_{\text{out}}H, \quad \mathbb{E}[\hat{\mathbf{x}}_{\text{out}} | \Theta] = R_{\text{out}}\mathbf{s} + A_{\text{out}}\boldsymbol{\mu}_0, \quad C_{x,\text{out}} = A_{\text{out}}N_{\text{PTC}}A_{\text{out}}^\top. \quad (9)$$

The diagonal  $D_{Q,\text{out}}$  contains only finite positive normalizations on  $\mathcal{S}_{\text{out}}$ , so its ordinary inverse is defined. The matrix  $A_{\text{out}}$  has no rows outside that set. Extending it to the full storage grid by writing zeros would be a representation convenience, not this scientific operator, response, or covariance.

Here  $N_{\text{PTC}}$  is a declared covariance of PTC-domain processed-sample error when such information is available, not the ensemble covariance of sky plus error. The state  $\Theta$  contains all selected or estimated calibration, alignment, membership, coefficients, response, geometry, identity, and effective-support facts. Equation (9) therefore makes a conditional statement on exactly the authorized rows. If any element of  $\Theta$  is estimated from the same data, an unconditional claim requires its joint uncertainty and cross-covariances.

When the required forward information is available, the response matrix  $R_{\text{out}} = A_{\text{out}}H$  describes what sky structure survives on the authorized output domain for its declared model, domain, and limitations. A stored kernel image is the realized response to a named unit-bearing template  $\boldsymbol{\tau}$  only if its sample-domain parent is  $\mathbf{k} = H\boldsymbol{\tau}$  and the stored map is  $A_{\text{out}}\mathbf{k} = R_{\text{out}}\boldsymbol{\tau}$  with the same exact row identity as signal. Without that identity it is a tracer, not the measured response. If the parent is missing or incomplete, response is explicitly unavailable or limited; no zero or identity array can stand in for it. That state does not invalidate the numerical map or prohibit later analysis. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity; it does not rewrite the original product's claim.

V0.1 fixes the signal-centering operator to  $L = I$ . SCI-MAP therefore does not remove a mean or a constant mode during coaddition. It preserves a constant admitted sample vector on the exact support-authorized output rows because numerator and denominator use identical coefficients. A stronger constant-sky statement additionally requires the forward operator to map that sky mode to the admitted sample constant. Neither result alone establishes point-source, integrated-flux, extended-source, edge, or absolute astrometric fidelity. Any absolute mode already removed upstream remains absent and must be carried in response and null-mode provenance.

### 3 Input classes and falsifiable response behavior

The same response equation yields distinct, testable input classes without granting them equal scientific meaning:

**Constant.**

Fixed positive coefficients preserve a constant admitted sample vector even when coverage and variance vary.

**Delta or basis pixel.**

The output is a support-row-restricted column of  $R_{\text{out}}$ , including beam, projection, preprocessing, and boundary effects.

**Point-source template.**

The output is  $R_{\text{out}}\tau_b$  on exactly the authorized rows. Peak, integrated, and fit amplitudes are different functionals.

**Resolved, gradient, and Fourier structure.**

The output is  $R_{\text{out}}\tau$ ; constant preservation says nothing about these modes.

**Variable coverage.**

Fixed-coefficient normalization may preserve a constant while covariance and nonconstant response vary spatially.

**Finite boundary.**

Truncation can preserve a local constant yet change a compact-source peak or integral. No wrap or clamp is authorized.

One-hot and fractional projection must remain distinguishable. With one-hot assignment and calibrated independent inverse-variance coefficients, ordinary normalized gridding coincides with the corresponding diagonal weighted mean. For fractional projection, the ordinary denominator contains  $G_{pi}$ , while the diagonal least-squares normal term contains  $G_{pi}^2$ . They are not the same estimator. This difference also creates cross-pixel covariance when one sample contributes to multiple pixels.

Response fidelity is therefore a validation layer, not a consequence of code executing the algebra. Delta, point-template, extended, Fourier, variable- coverage, and edge injections are preregistered in Appendix D; they are not run or claimed here.

## 4 Conditional uncertainty and statistical labels

For independent selected sample errors with true inverse-variance coefficients, the fractional-projection specialization is:

$$\text{Var}(\hat{x}_p \mid \Theta) = \frac{\sum_i G_{pi}^2 \omega_i}{(\sum_i G_{pi} \omega_i)^2}, \quad \text{Cov}(\hat{x}_p, \hat{x}_r \mid \Theta) = \frac{\sum_i G_{pi} G_{ri} \omega_i}{Q_p Q_r}, \quad p, r \in \mathcal{S}_{\text{out}}, \quad (10)$$

$$\phi_p = \frac{Q_p^2}{\sum_i G_{pi}^2 \omega_i}, \quad \phi_p = Q_p \iff \sum_i \omega_i (G_{pi}^2 - G_{pi}) = 0, \quad p \in \mathcal{S}_{\text{out}}, \quad (11)$$

Equation (11) states exactly why normalization is usually not precision. For  $0 \leq G_{pi} \leq 1$  and strictly positive coefficients, the equality is guaranteed by one-hot contributions. A single pixel receiving two coefficients  $G = 1/2$ ,  $\omega = 1$  has  $Q = 1$ , variance  $1/2$ , and formal marginal inverse variance  $\phi = 2$ . The off-diagonal term in Equation (10) remains whenever pixels share sample support.

Four uncertainty objects must not be conflated:

1.  $Q$ , an ordinary normalization with its declared gridding unit;
2.  $\phi = 1/[C_{x,\text{out}}]_{pp}$ , a conditional marginal inverse variance under its named assumptions;
3.  $C_{x,\text{out}}$  or  $C_{x,\text{out}}^+$ , the full conditional covariance or precision on the identifiable stochastic subspace of the exact authorized row domain; and
4. an empirical covariance, weight, or significance calibrated by NOI.

The bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. Missing correlations are named, not set to zero. Neither normalization, hits, exposure, nor diagnostic weight is silently promoted to covariance. Absence of a complete covariance model does not invalidate the map or prohibit later scientific analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity. The minimum persistence form remains an owner decision.

## 5 Eight facts, support, and raw validity

The raw bundle separates eight facts because they answer different questions:

| Fact                             | Meaning                                                                                                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Geometric projected incidences   | How many PTC-available, coordinate-valid candidate occurrences geometrically touched the pixel under the declared fractional counting convention. |
| Estimator-admitted contributions | How many occurrences passed every ordinary coefficient and contribution predicate.                                                                |
| Admitted observation count       | How many complete observation bundles contribute to a coadd pixel; not sample hits or attempted observations.                                     |
| Upstream-eligible exposure       | Named exposure over the source-available, coordinate-valid projected population, before MAP estimator-specific rejection.                         |
| Retained exposure                | The same named exposure over estimator-admitted membership and placement.                                                                         |
| Normalization support            | The numerical threshold predicate used to authorize ordinary division.                                                                            |
| Science-policy support           | The separately stricter adopted policy predicate.                                                                                                 |
| Final raw validity               | The authoritative conjunction of admitted identity, support, finite signal and required companions, and no product failure.                       |

No one fact is an alias for another. In particular, exposure is not precision, positive normalization is not final validity, and a finite payload is not evidence of scientific availability.

The adopted v0.1 threshold rule is policy authority, not a derived physical law. For the declared plane and policy population of finite positive  $Q_p$  values, sorted in ascending order, the admitted formula names  $c = \text{coverage\_cut}$ :

$$\mathcal{I}_{\Pi_{\text{eff}}} = \text{the declared support-policy population,} \quad (12)$$

$$\mathcal{P} = \{Q_p : \text{finite}(Q_p), Q_p > 0, p \in \mathcal{I}_{\Pi_{\text{eff}}}\}, \quad N = |\mathcal{P}|, \quad (13)$$

$$k = \left\lfloor \frac{\lfloor 0.75N \rfloor + N}{2} \right\rfloor, \quad Q_\star = \begin{cases} \mathcal{P}_{(k)}, & N > 0, \\ 0, & N = 0, \end{cases} \quad (14)$$

$$c = \text{coverage\_cut} \text{ (dimensionless scalar),} \quad \text{admit}_{\Pi_{\text{eff}}}(c) = 1, \quad (15)$$

$$T_{\text{norm}} = Q_\star c / 10, \quad T_{\text{sci}} = Q_\star c, \quad (16)$$

$$S_p^{\text{norm}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{norm}}], \quad (17)$$

$$S_p^{\text{sci}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{sci}}]. \quad (18)$$

Both predicates require the pixel's own explicit finite positive coefficient; the empty-population threshold of zero does not manufacture support. The operator row set  $\mathcal{S}_{\text{out}}$  is the exact intersection of the predicate row sets designated applicable by  $\Pi_{\text{eff}}$ ; it is not a full-grid mask with zero semantics.

The scientific owner has fixed  $c = \text{coverage\_cut}$  as a dimensionless support-policy scalar because  $Q_p$ ,  $Q_\star$ , and both thresholds are directly compared in the same coefficient unit. SCI-MAP-CI-001 records the resolved inconsistency and the bounded amendment to the canonical equations, REQ-031/032, and PRED-012. This disposition does not establish a numerical domain. Negative, zero, non-finite, and greater-than-one values, the recommended range, policy authority, and failure behavior remain SCI-MAP-OD-007. Threshold rationale and future change authority remain SCI-MAP-OD-001 and SCI-MAP-OD-002.

## 6 Observation coaddition on one common grid

Coaddition consumes immutable complete observation bundles, not independent planes. Admission is atomic: unit, estimand, response state, frame/WCS, shape, group/Stokes identity, support/validity policy, covariance status, parentage, and required companions are checked before any coadd-owned state changes.

The only v0.1 placement is centered integer embedding on the same grid. Observation and coadd row and column differences are nonnegative and even; their half-differences are the integer offsets, and the shifted reference pixels identify the same world coordinate. The lossless typed WCS is the authority. No fractional shift, interpolation, reprojection, wrapping, or implicit source recentering is allowed.

For the atomically admitted per-pixel set  $\mathcal{O}_p$ , every contributing observation row is already a member of that observation's support-authorized scientific domain. Let  $\mathcal{S}_{\text{out}}^c$  be the exact rows authorized by the effective coadd support policy. For finite positive observation coefficients  $u_{op}$ , the reused ordinary coadd is:

$$\mathcal{S}_{\text{out}}^c = \{p : \text{authorize}_{\Pi_{\text{eff}}^c}(p) = 1\}, \quad (19)$$

$$N_p^c = \sum_{o \in \mathcal{O}_p} u_{op} \hat{x}_{op}, \quad Q_p^c = \sum_{o \in \mathcal{O}_p} u_{op}, \quad (20)$$

$$\hat{s}_p = N_p^c / Q_p^c, \quad p \in \mathcal{S}_{\text{out}}^c, \quad (21)$$

$$J_{\text{out}}^c \mathbf{z} = (z_p)_{p \in \mathcal{S}_{\text{out}}^c}, \quad (22)$$

$$\hat{\mathbf{s}}_{\text{out}} = (\hat{s}_p)_{p \in \mathcal{S}_{\text{out}}^c} = B_{\text{out}} \hat{\mathbf{x}}_{\text{obs}}, \quad (23)$$

$$\hat{k}_p^c = \frac{\sum_{o \in \mathcal{O}_p} u_{op} \hat{k}_{op}}{Q_p^c}, \quad p \in \mathcal{S}_{\text{out}}^c. \quad (24)$$

$$C_{c,\text{out}} = B_{\text{out}} C_{\text{obs}} B_{\text{out}}^\top, \quad R_{c,\text{out}} = B_{\text{out}} T. \quad (25)$$

The coadd operator  $B_{\text{out}}$  has rows only in  $\mathcal{S}_{\text{out}}^c$ . Coadd accumulators or dense storage slots outside that set do not become a zero-valued scientific coadd. On its exact domain the coadd is a convex weighted mean. One admitted observation reproduces itself; unequal coefficients produce the corresponding weighted mean. Its normalization is not automatically precision. It is summed marginal inverse variance only when observations are unbiased unit-gain estimates of the same responded quantity, their errors are independent, and each  $u_{op}$  is the true marginal inverse variance. With cross-observation covariance, the exact conditional covariance retains those blocks on the same selected rows. A correlated-GLS benchmark may require negative combination coefficients, placing it outside this ordinary positive-coefficient contract.

Available response, retained exposure, and admitted observation count follow the same bundle admission and placement. A rejected observation changes no coadd state anywhere, even if a subset of its planes would otherwise appear compatible.



## 7 Identity, WCS, units, and lifecycle

A map cannot be reconstructed from its collection position. Paired PTC  $x/r$  components remain upstream detector coordinates and are carried only when a MAP-facing fact requires their identity; MAP does not recast  $r$  as a response. The minimum identity includes observation/coadd role; external observation or coadd ID; array and map group; network or occurrence-scoped detector binding where applicable; Stokes I; estimator; response state/version; signal unit; frame; full WCS; pixel shape/order and boundary policy; scientific MAP-policy identity; realized PTC parentage; and exact product identity. The policy must be scientifically identifiable, durable, and comparable, but this contract prescribes no JSON, hash, or sidecar serialization. Array ID, name, index, network, map group, map index, detector occurrence, and container position remain distinct.

Science maps use their declared equatorial J2000 TAN WCS with degree-valued spatial axes. A Pointing/OOF ordinary-map use, if the owner registers it, uses a declared AltAz tangent plane in arcseconds; its mode-specific meaning is outside SCI-MAP. FITS/WCS coordinates are one-based and in-memory row/column indices are zero-based.

Full-precision typed or sidecar WCS is the lossless admission and provenance authority. A physical FITS serialization may differ by at most 0.1 arcsec in maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. This is a representation tolerance, not authority for a different grid.

Requested state records the immutable accepted intent. Effective state records the executable supported plan. Observation-resolved state replaces all observation-specific arrays, mappings, geometry, coefficients, and provenance. Realized state records what actually ran, all accumulators and eight facts, products, offsets, representation states, cardinality, and failures. State flows one way; execution does not rewrite the request or inherit a prior observation's resolution.

## 8 Failure, downstream boundary, and immutable parentage

Failure semantics protect the scientific bundle from partial truth. An invalid candidate can be excluded at contribution scope; an unsupported pixel is raw-invalid; an incompatible observation is rejected atomically; an unrepresentable finite aggregate, coordinate/index conversion, missing required companion, or failed required publication propagates at its declared product scope before live state mutation. A completion marker cannot override a missing required output.

No downstream package receives a bare signal plane as a complete map. The boundary includes the estimator and normalization identity; response state; conditional covariance/formal-weight state and assumptions; eight facts; additive-bias and null-mode state; units; WCS; grouping; lifecycle; raw parentage; dependency versions; and failures. NOI may propagate a fixed operator but owns realization construction and empirical calibration. FLT owns filtered support, response, covariance, and validity. SRC, MODE, and BEAM own inference and fit uncertainty. FRUIT owns recurrence and iteration state.

Every later analysis is responsible for deciding which facts its own claim requires. An unavailable response prevents the original MAP bundle from making an unsupported response-dependent claim, but it does not prohibit analysis or invalidate other map claims. A significance analysis cannot relabel merely formal standardization; an absolute-level claim cannot ignore an unobservable offset. This contract defines no closed-world registry of all future uses. A downstream product may have local validity, but it cannot rewrite raw validity or sever the immutable raw-parent identity.

When an admitted noise realization is described as using the same operator, membership, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and coadd admission are fixed or identically prescribed. Recomputing them from randomized data creates a different estimator, regardless of matching array shapes or filenames.

## 9 Uncertainty, validity, validation, and claim layers

The contract deliberately separates four reasoning layers:

1. **Uncertainty layer:** conditional covariance and response are propagated through the fixed operators; omitted state uncertainty remains explicit.
2. **Validity layer:** contribution eligibility, numerical support, science-policy support, and final raw validity determine the exact rows on which the estimator is scientifically available; unsupported rows are absent, not zero-filled observations.
3. **Validation layer:** preregistered analytic fixtures, matrix comparisons, injections, covariance controls, representation round trips, and observational references test distinct claims.
4. **Readiness layer:** implementation evidence, operational reliability, data completeness, and release authority are assessed elsewhere.

| Claim layer                      | Status here           | Evidence needed before an achieved claim                                                                              |
|----------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Algebraic scientific contract    | Proposed author draft | Scientific-owner review and contract freeze; analytic predictions are specified, not executed.                        |
| Engineering conformance          | Not assessed          | Requirement-by-requirement inspection and tests against one exact candidate identity.                                 |
| Representation/response fidelity | Not assessed          | WCS round trips and delta, template, extended, variable-coverage, and edge injections with preregistered bounds.      |
| Observational performance        | Not assessed          | Independent astronomical amplitude, shape, position, covariance, and repeatability references where available.        |
| Production readiness             | Not assessed          | Complete required artifacts, failure/log audit, operational gates, owner release decision, and no disqualifying gaps. |

Floating reductions require a preregistered bound derived from operation count and conditioning. For a sum of  $n$  terms, a reference policy may use  $\gamma_n = n\epsilon/(1 - n\epsilon)$  and a fixed multiple such as the admitted core's  $8\gamma_n \sum_i |a_i|$ , frozen before observing results. Integer counts, identities, masks, support, offsets, and product cardinality remain exact. Tolerances are not relaxed after failure.

The contract's 25 predictions are listed formally in Appendix D. No validation run, implementation inspection, observational reduction, or production decision was performed during this scientific authorship stage.

## 10 Open owner decisions and scientific disposition

Nine decisions remain intentionally unresolved: (1) the physical rationale for the adopted support thresholds; (2) future threshold change authority and evidence; (3) the minimum response information that the original MAP product must provide for each response-dependent claim it makes; (4) the minimum covariance representation that must persist rather than remain lineage-resolvable; (5) whether physical observation-map publication is an abstract obligation beyond the effective v0.1 required-product rule; (6) whether Pointing/OOF ordinary arithmetic is a registered reuse of this operator; and (7) the admissible numeric domain, boundary-case dispositions, and failure behavior for the dimensionless  $c = \text{coverage\_cut}$ ; (8) authorized MAP projection classes, normalization, and boundaries; and (9) canonical-grid preparation and future reprojection/mosaicking ownership. Appendix [A](#) reproduces the exact questions and conservative interim behavior from the owner ledger. SCI-MAP-CI-001 is resolved and OD-007 is correspondingly narrowed without renumbering.

Pending those decisions, the scientific contract is coherent at its stated conditional boundary. It authorizes ordinary positive-coefficient normalized mapmaking and atomic centered-integer observation coaddition only. It does not establish implementation conformity, representation fidelity, observational performance, or production readiness.

## A Owner-decision register

This compact register is generated from `SCIENTIFIC_OWNER_DECISION_LEDGER.md`. The mechanical contract checker requires exact decision IDs, open statuses, decision prompts, and conservative interim dispositions to match before either PDF is accepted. Rendering this register does not answer or close an item.

| Decision/status               | Exact decision requested                                                                                                                                                                                                                                                     | Conservative draft behavior pending decision                                                                                                                                                                                                                                                           |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-MAP-OD-001<br><b>OPEN</b> | What physical rationale is authoritative for the adopted v0.1 normalization-support and science-policy-support threshold rule?                                                                                                                                               | State the formula exactly, label it adopted policy, and make no physical optimality, completeness, noise, or response claim from it.                                                                                                                                                                   |
| SCI-MAP-OD-002<br><b>OPEN</b> | Who may change the support-threshold rule after v0.1, and what evidence is required for a successor?                                                                                                                                                                         | Freeze the v0.1 formula and population identity; treat any altered multiplier, quantile/index, population, or predicate as a contract change.                                                                                                                                                          |
| SCI-MAP-OD-003<br><b>OPEN</b> | What response information must the original v0.1 MAP product provide for each response-dependent claim it makes, and which original product claims remain unavailable when response information is incomplete or unavailable?                                                | Report the actual response state, meaning, domain, and limits; fabricate neither zero nor identity response. Make no unsupported response-dependent claim. Permit later response or corrected-map products only with new versioned identity bound to the original MAP product and processing identity. |
| SCI-MAP-OD-004<br><b>OPEN</b> | What minimum covariance representation must persist in a raw bundle, and what may remain resolvable through lineage?                                                                                                                                                         | Report what uncertainty/covariance information exists, with meaning, domain, assumptions, limitations, omissions, and lineage. Do not promote normalization, hits, exposure, or diagnostic weight. Permit later estimates as new versioned products bound to the original.                             |
| SCI-MAP-OD-005<br><b>OPEN</b> | Beyond the adopted v0.1 effective required-output rule, is physical observation-map publication an abstract contract obligation whenever coaddition is requested?                                                                                                            | Preserve the v0.1 rule that every product designated required by the effective plan, including required observation products during coaddition, must publish successfully. Do not generalize this into a future universal storage mandate.                                                             |
| SCI-MAP-OD-006<br><b>OPEN</b> | Is Pointing/OOF ordinary-map arithmetic a registered use of the shared SCI-MAP v0.1 operator, with mode-specific interpretation remaining outside this package?                                                                                                              | State only that reuse is conditional on a fully conforming input bundle; grant no mode-specific fitting, astrometric, response, or production authority.                                                                                                                                               |
| SCI-MAP-OD-007<br><b>OPEN</b> | What numerical domain is authoritative for the dimensionless scalar <code>c=coverage_cut</code> , including the required disposition of negative, zero, non-finite, and greater-than-one values, and what failure behavior applies when a numerical state is not authorized? | Assert no general numerical domain. Require the exact numerical value to be explicitly admitted by the owner-authorized effective support policy; otherwise fail closed before constructing support-authorized output rows or mutating a required product.                                             |
| SCI-MAP-OD-008<br><b>OPEN</b> | Which sample-to-pixel projection classes are authorized for ordinary SCI-MAP v0.1, what normalization applies to MAP-owned <code>G_pi</code> (including whether and where <code>sum_p G_pi=1</code> is required), and how is boundary loss represented?                      | Require every product to declare its actual projection class, normalization, extent, and boundary convention. Permit the existing one-hot and fractional examples, but assert no unrecorded conservation property or additional projection class.                                                      |
| SCI-MAP-OD-009                |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                        |

| Decision/status | Exact decision requested                                                                                                                                                                                                         | Conservative draft behavior pending decision                                                                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OPEN</b>     | Is upstream crop or pad to a canonical common grid an authorized preparation for v0.1 observation coaddition, which package owns it, and which future package owns reprojection or mosaicking beyond centered-integer placement? | Reject every odd-difference or otherwise incompatible observation bundle. SCI-MAP performs no crop, pad, fractional shift, reprojection, interpolation, or mosaic; no other package is named as authorized until the owner decides. |

## B Requirement and crosswalk summary

The detailed 52-row requirement crosswalk and 25-row prediction crosswalk remain in `CROSSWALK.md`. This compact index makes the rationale’s traceability visible without duplicating the normative clause text or the detailed row table. Appendix C renders every canonical requirement, and Appendix D renders every prediction, directly from the shared authority.

| Authority range | Scientific surface                                                                        | Principal rationale / open gate                                  |
|-----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| REQ-001         | Version and revision identity                                                             | Cover; owner freeze                                              |
| REQ-002–010     | Capability, input, identity, projection, coefficients, lifecycle, and admission           | Sections 1, 7, 8; upstream producer facts                        |
| REQ-011–018     | Ordinary accumulators, exact support-row publication, response, and absolute mode         | Sections 1–3; OD-003                                             |
| REQ-019–024     | Row-selected covariance, formal weight, nuisance state, and statistical labels            | Sections 2, 4; OD-004                                            |
| REQ-025–035     | Eight facts, threshold policy, exact support rows, and raw validity                       | Section 5; OD-001, OD-002, OD-007                                |
| REQ-036–042     | Complete bundles, atomic centered-integer coadd, row-selected coadd                       | Sections 6, 8; OD-003, OD-004, OD-007                            |
| REQ-043–048     | response/covariance, and GLS boundary                                                     |                                                                  |
| REQ-043–048     | Product identity, WCS, provenance, failure, and publication                               | Sections 7–8; OD-005, OD-006                                     |
| REQ-049–052     | Immutable raw parent, consumer boundary, fixed realization operator, and claim separation | Sections 8–9; OD-003, OD-004, OD-006, OD-007                     |
| PRED-001–025    | Analytic consequences and conformance/representation challenges                           | Sections 3, 9; Appendix D; not executed                          |
| OD-001–009      | Exact unresolved scientific-owner questions                                               | Section 10; Appendix A; all OPEN                                 |
| SCI-MAP-CI-001  | Resolved dimensionless classification and bounded amendment                               | Section 5; equations, REQ-031/032, and PRED-012 amended in place |

## C Canonical requirements

This appendix is rendered directly from the shared canonical LaTeX authority. Requirement identifiers are stable within SCI-MAP v0.1. Editing prose in this document does not silently change them.

**SCI-MAP-REQ-001 – Versioned authority.** The scientific contract version shall be v0.1. The rendered document revision shall be recorded separately as r0.4; changing layout or exposition alone shall not be represented as a new scientific contract version.

**SCI-MAP-REQ-002 – Capability and realized PTC input.** The ordinary input shall be a realized SCI-PTC MAP-facing product derived from calibrated ordinary-**xs** Stokes-I samples, with exact sample-by-detector membership and active signal unit **mJy/beam**. MAP shall have no CAL-to-MAP fallback: an identity-like PTC configuration remains a realized PTC product, and absence of the required PTC product means that MAP has no ordinary input.

**SCI-MAP-REQ-003 – Explicit identity.** Every candidate contribution shall carry explicit observation, array, network when applicable, map-group, Stokes, input-column, and occurrence/product-scoped detector-acquisition identity. Container position, row equality, or a local key alone shall not establish scientific identity.

**SCI-MAP-REQ-004 – Producer facts and MAP admission.** SCI-MAP shall consume without reinterpretation PTC-owned product identity, availability, producer-local validity, and MAP-facing coefficients, together with other producer-owned facts. MAP contribution shall require the realized PTC output to be available and every applicable MAP-owned projection, coefficient, boundary, finite-value, support, and use condition to pass. VAL may register or evaluate a named rule but shall not author either producer or MAP policy.

**SCI-MAP-REQ-005 – Coordinate and projection authority.** ALIGN/AST shall own sample coordinates, time/frame identity, astrometry, and coordinate validity. MAP shall own the target grid and its declared sample-to-map relation  $G_{pi}$ , projection class, normalization, extent, and boundary convention. MAP shall not choose or repair pointing, interpolate a missing coordinate, reinterpret coordinate validity, wrap, clamp, or invent an out-of-grid placement.

**SCI-MAP-REQ-006 – Ordinary coefficient status.** Each admitted ordinary analysis coefficient shall be finite and strictly positive, with declared unit, normalization scope, lifecycle, applied factors, and statistical status. Zero is noncontributing; negative and non-finite ordinary coefficients are invalid.

**SCI-MAP-REQ-007 – Exposure meaning.** Every exposure contribution shall be finite and nonnegative in a named unit and accounting convention. Detector-seconds shall not be labeled wall-clock integration time without an explicit multiplicity normalization.

**SCI-MAP-REQ-008 – Response information state.** MAP shall report the response information it actually provides, including meaning, domain, parentage, normalization, units, and limitations, or an explicit typed unavailable state. Missing or incomplete response shall not invalidate an otherwise valid map, fabricate an identity response, or prohibit later scientific analysis. A later simulated response or response-corrected map shall be a new versioned product bound to the original MAP product and processing identity.

**SCI-MAP-REQ-009 – Immutable plan and lifecycle.** The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

**SCI-MAP-REQ-010 – Contribution predicate.** For each pixel, the estimator contribution set shall require a realized PTC output available under the producer's exact declared identity and scope, plus finite signal, finite positive effective coefficient, admitted MAP projection and boundary, support/use policy, and every required MAP companion predicate before payload accumulation. No universal occurrence-to-product or product-to-occurrence failure hierarchy shall be inferred.

**SCI-MAP-REQ-011 – Ordinary accumulators.** On the fixed contribution set, SCI-MAP shall accumulate the pre-normalized signal numerator  $N_p = \sum_i a_{pi} x_i$ , normalization  $Q_p = \sum_i a_{pi}$ , and, when available, response numerator  $K_p = \sum_i a_{pi} k_i$ , where  $a_{pi} = G_{pi} \omega_i > 0$ .

**SCI-MAP-REQ-012 – Normalized publication.** The scientific normalized output vector shall contain exactly the rows  $p \in \mathcal{S}_{\text{out}}$  authorized by the applicable effective support policy. On those rows a finite  $Q_p > 0$  gives



$\hat{x}_p = N_p/Q_p$  and an available response companion gives  $\hat{k}_p = K_p/Q_p$ . Accumulators or storage slots may exist elsewhere, but no unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero sky.

**SCI-MAP-REQ-013 – Accumulator interpretation.** The numerator is a pre-normalized accumulator and  $Q_p$  is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

**SCI-MAP-REQ-014 – Constant preservation.** With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

**SCI-MAP-REQ-015 – Projection-specific estimand.** One-hot assignment and fractional projection shall be identified explicitly. For fractional projection, normalized gridding shall not be mislabeled diagonal weighted least squares, and  $Q_p$  shall not be promoted to formal precision by analogy.

**SCI-MAP-REQ-016 – Same operator for linear companions.** Signal, available response/kernel tracer, and declared linear realizations shall use the same admitted membership, geometric placement, and exact support-authorized output-row selection; retained exposure and coadd observation count shall use the same membership and placement wherever their definitions require them.

**SCI-MAP-REQ-017 – Realized response claim.** A stored kernel may be called the realized response to template  $\tau$  only when its sample parent satisfies  $\mathbf{k} = H\tau$  and the support-restricted map product satisfies  $\hat{\mathbf{k}}_{\text{out}} = A_{\text{out}}H\tau = R_{\text{out}}\tau$  on exactly the authorized rows. Otherwise it shall be labeled only a tracer or unavailable.

**SCI-MAP-REQ-018 – Absolute-offset mode.** SCI-MAP v0.1 shall apply no signal-centering operation:  $L = I$ . Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

**SCI-MAP-REQ-019 – Conditional covariance.** When a declared PTC-domain covariance  $N_{\text{PTC}}$  is available for the asserted domain and fixed complete state  $\Theta$ , MAP shall propagate it on the exact support-authorized row set as  $C_{x,\text{out}} = A_{\text{out}}N_{\text{PTC}}A_{\text{out}}^T$ , retaining consequential cross-pixel terms. The equation does not require MAP to provide a complete covariance model when its input is absent or incomplete, and no zero-filled extension is scientific covariance.

**SCI-MAP-REQ-020 – When normalization is precision.** The normalization may be called marginal inverse variance only when coefficients are calibrated true inverse variances, selected sample errors are independent for the asserted domain, and the projection satisfies  $\sum_i \omega_i (G_{pi}^2 - G_{pi}) = 0$ ; one-hot assignment is the standard sufficient case.

**SCI-MAP-REQ-021 – Formal diagonal weight.** When its assumptions hold on an authorized output row, the formal marginal inverse variance shall be  $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \omega_i$ . It shall remain distinct from  $Q_p$ , from the support-restricted full precision matrix  $C_{x,\text{out}}^+$ , and from empirical statistical weight.

**SCI-MAP-REQ-022 – Covariance and nuisance disclosure.** The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

**SCI-MAP-REQ-023 – Data-derived state.** If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

**SCI-MAP-REQ-024 – No significance promotion.** Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. Empirical calibration remains NOI-owned.

**SCI-MAP-REQ-025 – Eight distinct facts.** Every raw bundle shall separately represent geometric projected incidences, estimator-admitted contributions, admitted observation count for coadds, upstream-eligible exposure,

retained exposure, numerical normalization support, science-policy support, and authoritative final raw science validity. No alias shall become authority for another fact.

**SCI-MAP-REQ-026 – Geometric incidences.** Geometric projected incidences shall count upstream-eligible entries whose declared projection touches a pixel, using an explicit fractional-projection counting convention, regardless of whether a positive ordinary coefficient is later admitted.

**SCI-MAP-REQ-027 – Estimator contributions.** Estimator-admitted contributions shall count exactly the entries in the ordinary contribution set and shall exclude zero, negative, non-finite, flagged, and out-of-bounds candidates.

**SCI-MAP-REQ-028 – Admitted observation count.** For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

**SCI-MAP-REQ-029 – Upstream-eligible exposure.** Upstream-eligible exposure shall sum the named exposure measure over the externally eligible projected population, before estimator-specific coefficient rejection, according to its declared exposure projection.

**SCI-MAP-REQ-030 – Retained exposure.** Retained exposure shall sum the same named exposure measure over estimator-admitted membership and placement; it shall not be labeled precision.

**SCI-MAP-REQ-031 – Adopted threshold selector.** For each declared support-policy population, sort its finite strictly positive normalization coefficients  $Q_p$ ; for count  $N > 0$ , select zero-based  $k = \lfloor ([0.75N] + N)/2 \rfloor$  and  $Q_\star = Q_{(k)}$ ; for empty input set  $Q_\star = 0$ . The exact population identity and exact dimensionless support-policy scalar  $c = \text{coverage\_cut}$  shall be recorded. The effective support policy shall explicitly admit that exact numerical  $c$  state before support resolution; otherwise the required product shall fail closed.

**SCI-MAP-REQ-032 – Separate support predicates.** Only after the effective support policy explicitly admits the exact value of the dimensionless scalar  $c$ , numerical normalization support shall require an explicit finite positive  $Q_p \geq Q_\star c/10$ , and science-policy support shall independently require an explicit finite positive  $Q_p \geq Q_\star c$ . This contract asserts no general numeric domain for  $c$ . Every non-authorized numerical state fails closed before a support row set is constructed. Neither support state shall establish final validity or precision.

**SCI-MAP-REQ-033 – Authoritative raw validity.** One MAP raw science-validity state shall require membership in the exact support-authorized output-row set, admitted PTC parent identity, applicable MAP support predicates, finite normalized signal, and absence of an applicable declared-scope MAP product failure. Response or covariance may be explicitly incomplete or unavailable unless a particular MAP-authored claim requires it. A retained slot, finite payload, positive coefficient, exposure, hit count, support plane, or compatibility alias alone shall not make a pixel valid.

**SCI-MAP-REQ-034 – Screen before arithmetic.** Invalid contributions shall be excluded before their numerical payload is evaluated. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

**SCI-MAP-REQ-035 – Exceptional contribution states.** Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

**SCI-MAP-REQ-036 – Complete observation bundle.** An observation map shall be one immutable ordered bundle containing signal, numerator, normalization identity, honest response state, honest uncertainty/covariance state, all eight facts where applicable, units, frame/WCS, shape/indexing, estimator, lifecycle, realized PTC parentage, and exact product identity. Explicitly incomplete or unavailable response/covariance states are complete disclosures, not automatically invalid bundles.

**SCI-MAP-REQ-037 – Atomic coadd admission.** A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

**SCI-MAP-REQ-038 – Centered integer common grid.** Observation and coadd shapes shall differ by non-negative even row and column counts, with the corresponding integer offset mapping their reference pixels to the

same world coordinate. Fractional shift, reprojection, interpolation, wrapping, implicit recentering, and a different sky grid are forbidden.

**SCI-MAP-REQ-039 – Ordinary coadd estimator.** For atomically admitted bundles and finite positive observation coefficients  $u_{op}$ , the coadd shall accumulate  $N_p^c = \sum_o u_{op} \hat{x}_{op}$  and  $Q_p^c = \sum_o u_{op}$ , then publish  $\hat{s}_p = N_p^c / Q_p^c$  only for rows in the exact effective coadd support set  $\mathcal{S}_{out}^c$ . No unsupported coadd row or zero-filled substitute is a scientific output.

**SCI-MAP-REQ-040 – Coadd companions.** Available response/kernel shall use the same atomic admission, centered-integer placement, and exact support-authorized output-row selection as coadd signal. Retained exposure and admitted observation count shall use the same admission and placement; the count increments once per admitted complete bundle at each applicable pixel.

**SCI-MAP-REQ-041 – Coadd covariance and precision.** For fixed admission and support-restricted coadd operator  $B_{out}$ , conditional covariance shall be  $C_{c,out} = B_{out} C_{obs} B_{out}^T$ , including consequential cross-observation blocks among authorized rows. Summed observation coefficients equal coadd precision only for independent unbiased maps of the same unit-gain estimand weighted by their true marginal inverse variances.

**SCI-MAP-REQ-042 – Correlated-GLS boundary.** A correlated minimum-variance solution using  $K_p^{-1} \mathbf{1}$  may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

**SCI-MAP-REQ-043 – Minimum product identity.** Every observation and coadd product shall identify role, external observation or coadd identity, array and group, network/detector occurrence when applicable, Stokes I, estimator, response and covariance disclosure/version, signal unit, frame, WCS, pixel shape/order, boundary policy, MAP policy identity, realized PTC parentage, and product identity. Scientific policy identity shall be durable and comparable without this contract prescribing JSON, hashes, or a sidecar format.

**SCI-MAP-REQ-044 – Frames and units.** Science products shall use their declared equatorial J2000 TAN WCS with degree-valued spatial axes; Pointing/OOF registered uses, if approved, shall use a declared AltAz tangent plane in arcseconds. Signal and a unit-bearing realized response carry mJy/beam; covariance carries its square and a conditionally valid formal weight its inverse square.

**SCI-MAP-REQ-045 – WCS and index persistence.** Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

**SCI-MAP-REQ-046 – Realized provenance.** Realized state shall record executed observations and groups, admitted contributions and bundles, accumulators, exact observation and coadd support-authorized output-row sets, all eight facts, response and covariance representation states, centered-integer offsets, completed required products, failures, and the exact requested/effective/resolved parents.

**SCI-MAP-REQ-047 – Failure scope and atomicity.** An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

**SCI-MAP-REQ-048 – Required observation products.** A complete observation bundle remains required when coaddition is enabled, and every product designated required by the effective v0.1 plan shall be published successfully before completion is asserted. Whether future abstract contracts require physical observation-map publication independent of an effective product policy remains an owner decision.

**SCI-MAP-REQ-049 – Immutable raw parent.** Downstream transformations shall preserve resolvable raw bundle identity and the authoritative raw validity state. They shall not rewrite raw validity or promote a raw-invalid pixel through a filtered or consumer-local validity alias.

**SCI-MAP-REQ-050 – Complete consumer boundary.** No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include estimator/normalization identity, honest response and uncertainty/covariance disclosures, eight facts, additive-bias/null-mode state, units, frame/WCS, shape/indexing, grouping, lifecycle, parentage, dependency versions, and declared-scope failures. Each later analysis determines which available facts its own claim requires; this contract shall not impose a closed-world consumer/use registry.

**SCI-MAP-REQ-051 – Same fixed noise operator.** An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

**SCI-MAP-REQ-052 – Method and claim separation.** This authority shall not be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

## D Falsifiable predictions

These predictions are preregistered contract consequences or conformance challenges. Their presence is not validation evidence.

**SCI-MAP-PRED-001 – Uniform constant.** For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized raw-valid output row; numerator and normalization scale together, and no unsupported scientific row is created. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-002 – Unequally weighted constant.** For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-003 – One-pixel hand calculation.** For an authorized output row with admitted values  $x_i$  and coefficients  $a_i > 0$ , the exact expected triplet is  $(N, Q, \hat{x}) = (\sum a_i x_i, \sum a_i, \sum a_i x_i / \sum a_i)$ ; for an unauthorized row only declared accumulators or storage state may remain, never a normalized scientific zero. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-004 – Fractional split.** One sample split with  $G_{p1} = G_{r1} = 1/2$  contributes to both pixels, creates cross-pixel covariance, and distinguishes normalized gridding from diagonal WLS. Two equal independent split samples give  $Q = 1$ , variance  $1/2$ , and  $\phi = 2$  per affected pixel. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-005 – Delta input.** A basis-pixel sky input produces the corresponding support-row-restricted column of  $R_{\text{out}} = A_{\text{out}}H$ , including projection and boundary truncation; it need not be a delta in the output map, and no value is predicted outside the authorized rows. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-006 – Point-source template.** A declared point-source template  $\tau_b$  produces  $R_{\text{out}}\tau_b$  on exactly the support-authorized rows; peak, integrated, and fitted amplitudes are distinct functionals and need separate criteria. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-007 – Resolved and Fourier inputs.** Resolved disks, gradients, and Fourier modes produce their explicit  $R_{\text{out}}\tau$  on the authorized row set; a constant-input pass does not predict preservation of these modes. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-008 – Variable coverage.** Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-009 – Finite boundary.** At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-010 – Coefficient states.** Zero coefficients add no estimator contribution; low positive coefficients follow the adopted support policy; negative and non-finite ordinary coefficients are invalid and cause-specific. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-011 – Eligibility and non-finite states.** Flagged, non-finite, and out-of-bounds candidates do not contaminate any accumulator because they are excluded before payload evaluation; zero or failed support produces no scientific output row rather than measured zero. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-012 – Threshold selector.** For each predeclared positive finite coefficient population, the selected zero-based index is exactly  $\lfloor ([0.75N] + N)/2 \rfloor$ ; empty input yields threshold zero, but no pixel gains support without its own finite positive coefficient. The dimensionless identity of  $c$  is fixed; any exact numerical  $c$  value not explicitly admitted by the owner-authorized effective policy fails closed before a support row set is constructed, and no numeric domain is inferred. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-013 – Missing response.** A missing response parent yields a typed unavailable response state, never a fabricated zero or identity response. The original MAP bundle remains usable for claims that do not require response; any later simulated response or corrected map has a new versioned identity bound to that bundle. *Primary claim layer: Engineering conformance.*



**SCI-MAP-PRED-014 – Atomic rejection.** A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-015 – One-observation coadd.** A coadd containing one admitted observation with positive coefficient reproduces that observation’s signal and available response exactly on the corresponding centered-integer rows authorized by both the observation and coadd support policies; it creates no unsupported scientific row. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-016 – Unequal observation coefficients.** On each support-authorized coadd row, two compatible observations produce the positive-coefficient weighted mean and the summed normalization; the result remains inside the convex hull of their finite values, while an unauthorized row has no normalized coadd value. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-017 – Missing or invalid observation.** An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-018 – Centered integer placement.** Compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-019 – Cross-observation covariance.** For two maps with variances  $\sigma_1^2, \sigma_2^2$  and covariance  $c$ , the minimum-variance coefficients depend on  $c$ ; summing marginal normalizations is not precision unless the required independence and calibration conditions hold. Covariance comparison uses the exact support-authorized observation and coadd rows. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-020 – Fixed noise operator.** A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient  $A_{\text{out}}$  and  $B_{\text{out}}$  as the signal path; recomputed state exposes a named difference. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-021 – Sequential and parallel reduction.** Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-022 – No false significance.** A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-023 – No validity promotion.** A downstream product may have its own local validity but cannot change an inherited raw-invalid pixel to raw-valid or sever its raw-parent identity. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-024 – No ordinary predicate for JINC.** A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.1. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-025 – Aggregate and index failure.** An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion. *Primary claim layer: Engineering conformance.*

## E Product vocabulary

| Product                  | Unit                                 | Contract meaning                                                                                                                                            |
|--------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal numerator         | coefficient times<br><b>mJy/beam</b> | Pre-normalized accumulator; not a sky estimator.                                                                                                            |
| Normalization            | declared gridding<br>unit            | Positive denominator; precision only under Requirements 020–021.                                                                                            |
| Normalized signal        | <b>mJy/beam</b>                      | Conditional ordinary map estimand whose vector contains exactly the effective-policy-authorized output rows; no unsupported zero-filled semantic extension. |
| Formal diagonal weight   | inverse square signal<br>unit        | Conditional marginal inverse variance under named assumptions; not empirical significance.                                                                  |
| Conditional covariance   | square signal unit                   | Full fixed-state noise propagation on the exact scientific output-row domain, with omissions and representation status named.                               |
| Response/kernel          | <b>mJy/beam</b> when<br>unit-bearing | Realized template response only when the sample-parent identity is proven; otherwise tracer or unavailable.                                                 |
| Geometric/estimator hits | integer                              | Counts of declared populations, not coverage, precision, or validity.                                                                                       |
| Exposure facts           | named<br>time/accounting<br>unit     | Upstream-eligible and retained accounting measures, not precision.                                                                                          |
| Support facts            | logical                              | Separate normalization and science-policy predicates; their effective conjunction defines output-row membership.                                            |
| Raw validity             | logical                              | Authoritative consumer-eligibility state for the raw bundle, preserving required identity and companion finiteness.                                         |

## F Admitted source register and independence statement

The scientific author used only the following admitted sources:

| Source                    | Exact admitted identity                                                                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope Brief               | SCOPE_BRIEF.md<br>SHA-256 e2a9eb51edb5956191813b4cbbd23866e875d52cdf89cd8b6c272988b4f26674.                                                                                                     |
| Supersession Cover        | AUTHOR_SUPERSESSION_COVER.md<br>SHA-256 8ea283525f18199d9760c3f672d145d71f0db87b320ab2e88a5c6635ef3d4aa0.                                                                                       |
| Frozen core               | Exact Git object c28f18ed089657dae278caba2d6d6d65c7ec72f4:<br>doc/audits/packages/SCI-MAP-001_INDEPENDENT_CORE.tex<br>SHA-256 13dd5922bd492e381afcc3b015284216dde1ccc2199ece3d070ee577c7324381. |
| Conventions and ownership | AUTHOR_CONVENTIONS_AND_OWNERSHIP.md<br><br>SHA-256 2d478cb6c5e897308d19614b8b01663318744971850c67459f84c7ddcd57c5c9.                                                                            |
| Owner handoff correction  | CHANGE_LOG_R0.4.md, recording the 2026-08-26 owner-authorized bounded PTC-to-MAP correction.                                                                                                    |

No implementation, test, validation, reduction, production-status record, MAP-002/MAP-003 evidence, or web source entered this authorship. The bounded handoff review supplied only the owner-authorized authority corrections recorded above. The frozen ordinary derivation was consolidated rather than rederived for novelty.